wtchtwr

AI-Native Analytics Engine for the Short-Term Rental (STR) Industry

Prepared for:

GWSB AI Case Competition

Prepared by:

Group 13 - MSBA

Date

05 December 2025

GW Business

The Short Term Rental (STR) Market

hometeam VACATION RENTALS

STAYTERRA

stayflexi

ukio

+5.23%

increase in average daily rates (ADR) for upscale listings [1]

76%

of STR managers surveyed emphasized the need for smarter pricing and operational efficiencies [2]

62%

of respondents believe that AI offers a competitive advantage in the marketplace [3]

Market Pain Points

01.

Fragmented Data Sources

No centralized system to track metrics and guide portfolio decisions as data is scattered across teams and systems.

02.

Weak Market & Competition Analysis

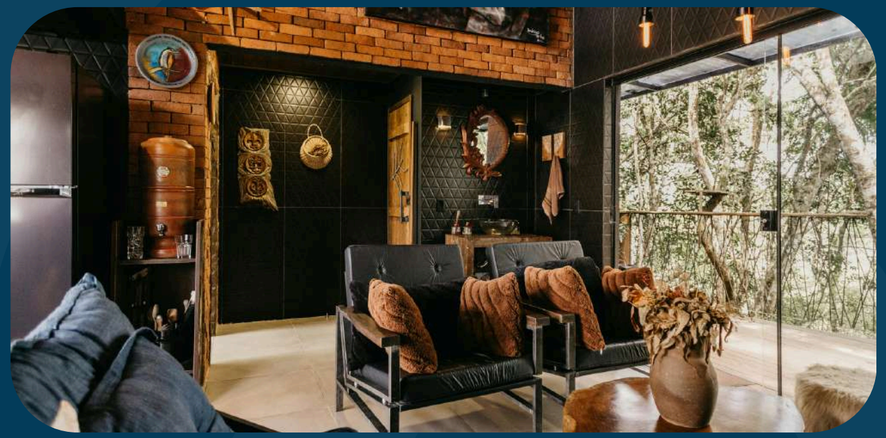
Disconnected tools make it hard for teams to benchmark performance get a broader picture of the market.

03.

Lack of Data Backed Strategy

Fragmented systems and weak market analysis make it harder for organizations to leverage data in their day-to-day operations.

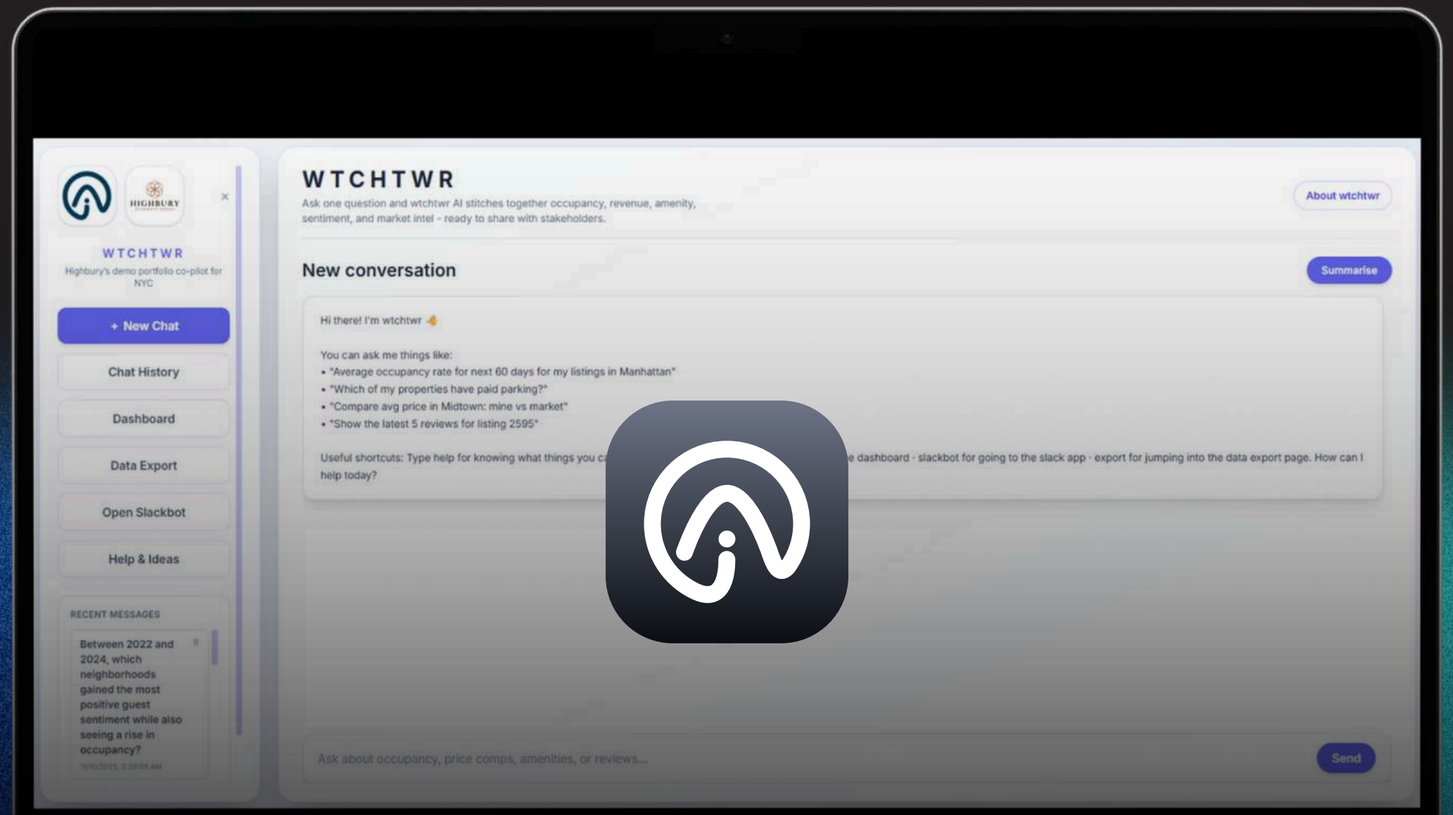
The Problem We Solve



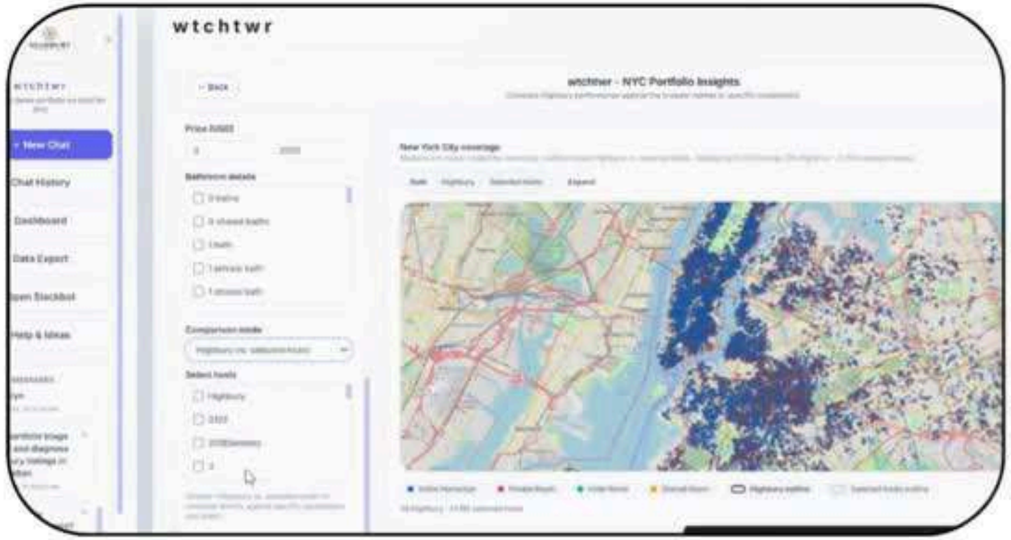
STR managers are flying blind. Their data is scattered across channels and tools, making it hard to get clear, verifiable answers about how their portfolio is performing versus the market.

We fix this by giving them a centralized**, easy-to-use BI "watchtower"** for portfolio and competitive performance.

Introducing the **wtchtwr** Platform



wtchtwr is an AI portfolio intelligence co-pilot combining structured analytics, review intelligence, autonomous triage and scouting agents built to help STR operators make smarter decisions, faster.



Don’t just analyse your portfolio,
benchmark it against the market

Key Features

Luxury STR Business Strategy, **Reimagined**

Portfolio Triage Engine

Auto-flags fix-first listings and underpriced winners to prioritize action

Expansion Scout

Reveals high-demand, under-served pockets for smart expansion bets

KPI & Performance Analysis

Spot portfolio winners/underperformers vs market at a glance

Pricing & Revenue Intelligence

Find under/overpriced listings and where price moves matter most

Sentiment & Review Insights

Turn reviews into clear themes explaining performance gaps

Explainable AI

Exposes the data, SQL, and reviews behind each recommendation for a clear audit trail

wtchtwr's Impact

Data-Driven Portfolio Decisions

Expansion Scout + BI insights highlight where to expand, prune, and reinvest so portfolio moves are smarter, faster, and backed by evidence

Operational Efficiency Boost

Email/Slack workflows + sentiment-linked reviews-to-KPIs so teams quickly fix concrete property issues instead of guessing

Revenue Upside

Triage mode flags underpriced winners and fix-first listings so ADR moves are focused where they drive the most revenue

Time & Cost Savings

Standard BI report cost reduced by ~86% per deep dive (~$330 saved per report at $55/hr) [4]

Time saved

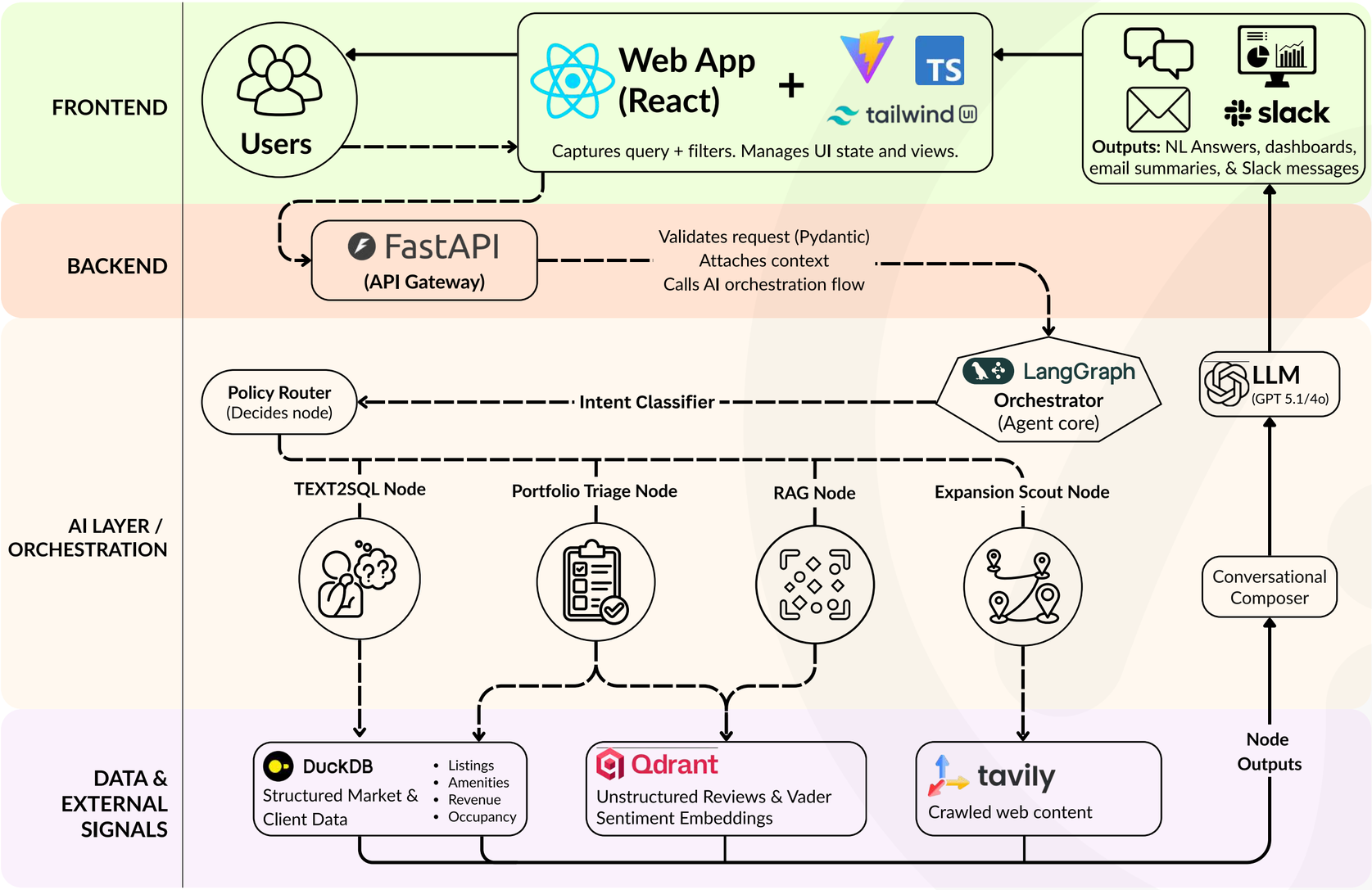
~7 hours → ~1 hour

Cost saved

$385 → $55

~86% **reduction in both time and costs** [4]

Tech Architecture



The Competition?

	wtchtwr	AIRDNA	AllTheRooms	Wheelhouse	ChatGPT
Purpose built	✓	✗	✗	✗	✗
Natural-language analytics on proprietary schema	✓	✗	✗	✗	✗
Review insights (RAG)	✓	✗	✓	✗	✗
Transparent SQL lineage	✓	✗	✗	✗	✗
Open-source market data blending	✓	✗	✗	✗	✓
Built in Slack / Email integration	✓	✗	✗	✗	✗
Enterprise data privacy	✓	✓	✗	✗	✗

Fictional Demo Client

HIGHBURY
PROPERTY GROUP

38
Listings in NYC
Focussed in Manhattan

~$150M
Real Estate
Assets Under Management

Friction Points:

Fragmented Data Systems, Operating in a limited market, Lack of competition analysis, Manual & Inefficient Workflows, Lack of Data-Backed Strategy

Value Delivered to Customer

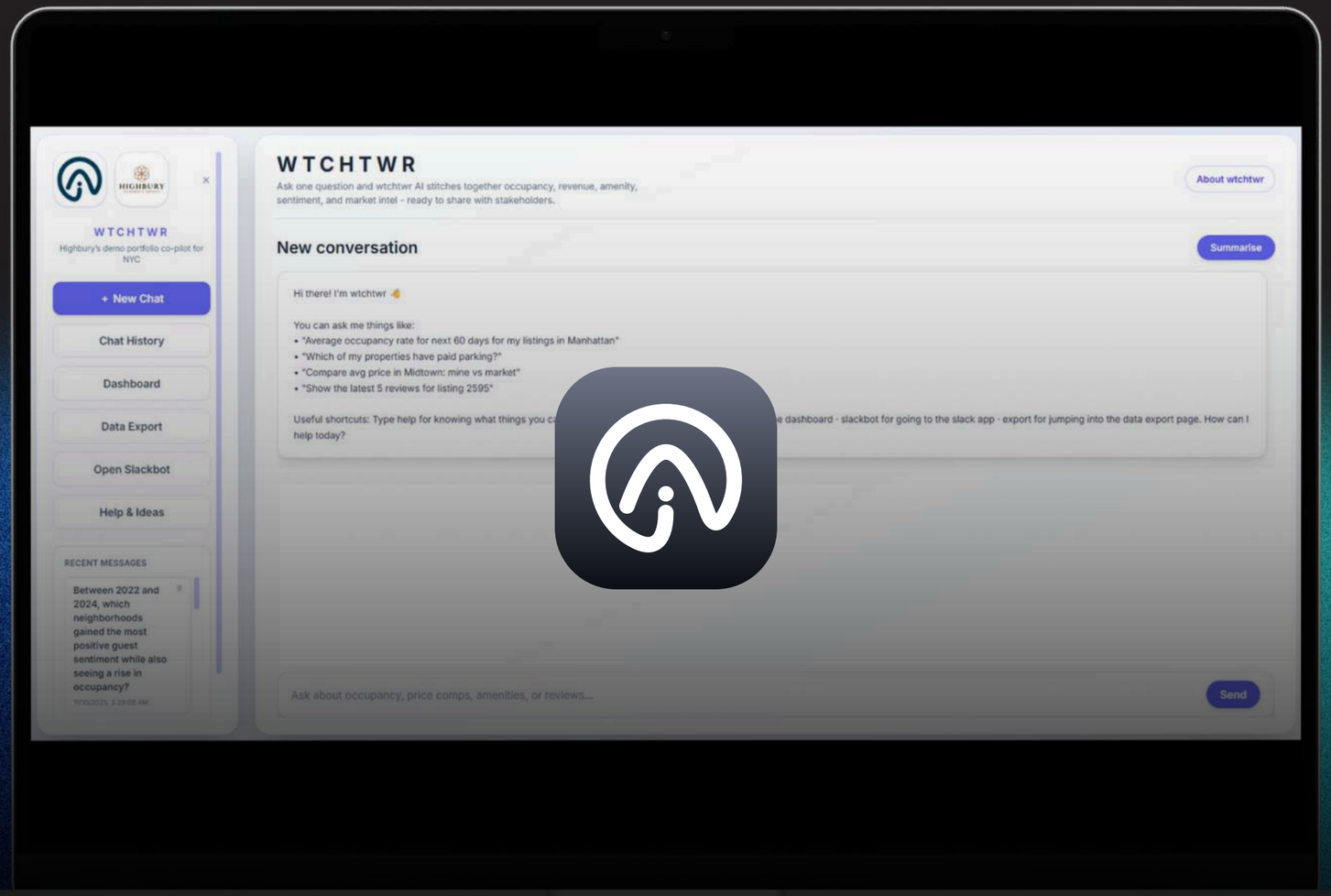
Competitive Market Intelligence

Operational Efficiency & Automation

Data-Driven Portfolio Decisions

Market Expansion Intelligence

Demo


WTCHTWR
Highbury's demo portfolio co-pilot for NYC
+ New Chat
Chat History
Dashboard
Data Export
Open Slackbot
Help & Ideas
RECENT MESSAGES
Between 2022 and 2024, which neighborhoods gained the most positive guest sentiment while also seeing a rise in occupancy?
WTCHTWR
Ask one question and wtchtwr AI stitches together occupancy, revenue, amenity, sentiment, and market intel - ready to share with stakeholders.
About wtchtwr
New conversation
Summarise
Hi there! I'm wtchtwr
You can ask me things like:
• "Average occupancy rate for next 60 days for my listings in Manhattan"
• "Which of my properties have paid parking?"
• "Compare avg price in Midtown: mine vs market"
• "Show the latest 5 reviews for listing 2595"
Useful shortcuts: Type help for knowing what things you c
e dashboard · slackbot for going to the slack app · export for jumping into the data export page. How can I help today?
Ask about occupancy, price comps, amenities, or reviews...
Send

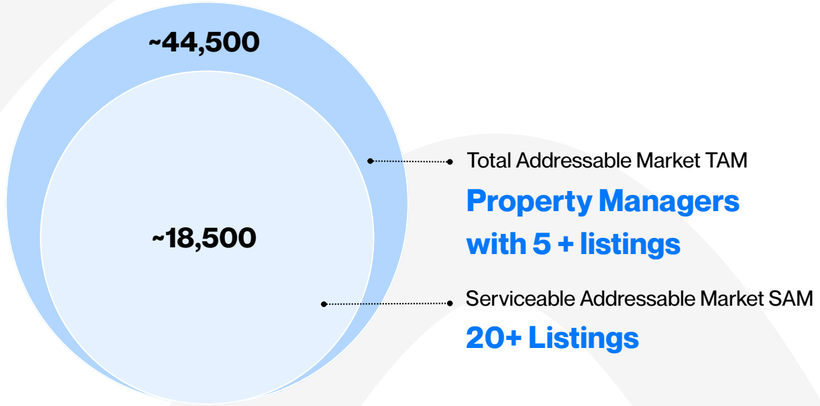
Market Opportunity & Business Model

+3.57%

Expected Market Growth Rate from 2025 to 2034[5]

24%

Market Share by North American STR Managers[5]



Subscription-Based SaaS - Tiered monthly plans based on portfolio size.

Starter	Growth	Pro	Enterprise
0-50 Listings	51-200 Listings	201-500 Listings	500+ Listings
$150/ Month	$450/ Month	$900/ Month	From $2,000/ Month

$350K

Seed Investment

85–88%

Gross Margin [6]

~3-4x

Life Time Value/Cusomer Acquisition Cost [6]

<7

months

CAC Payback [6]

Risk Considerations & Guardrails

Bias & Accuracy

Model outputs can mirror biased market/review patterns

Monitor skew and keep humans in control of key decisions

Data Privacy & Personally Identifiable Information

Guest and owner data can contain sensitive identifiers

Minimize, mask, and restrict access to guest/owner identifiers and avoid sending raw PII to external APIs or LLMs

Model Hallucinations & Reliability

LLMs can generate confident but incorrect explanations or recommendations

All narratives are grounded in KPIs and retrieved reviews, logged for traceability, and be subject to human review before action

Future Scalability

Upgrade intent classifier
to a learned model (ML/RL) instead of simple keyword matching

Expansion Scout 2.0
Expand scout to listing level with richer demand signals (events, seasonality, travel data)

Deploy full agentic capabilities
so the system can take actions (e.g., update listing prices via APIs)

Add built-in risk & compliance monitoring
by integrating open data (code violations, permits, complaints) into triage and expansion decisions

Add multi-platform integrations
across Airbnb, Vrbo, Booking.com, and other STR and Hotel channels

Meet the Founding Team

Gokul

Technical

Manali

Documentation

Sibtain

Technical

Savera

Client Communications

Bhuwan

Project Management

Your market **watchtower**

wtchtwr

GW Business

References

[1] Stayfi (2025). Vacation Rental Statistics, Data & Trends 2025
https://stayfi.com/vrm-insider/2025/07/28/vacation-rental-statistics/

[2] Rent Responsibly (2024). 2024 Vacation Rental Stats Roundup
https://www.rentresponsibly.org/2024-vacation-rental-stats-roundup/

[3] Hostaway (2024). How Many Short-Term Property Managers Are Using Artificial Intelligence
https://www.hostaway.com/blog/ai-short-term-rental-report/

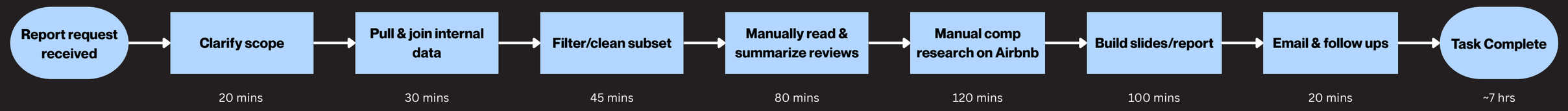
[5] Precedence Research (2025). Vacation Rental Market Size, Share & Trends 2025–2034
https://www.precedenceresearch.com/vacation-rental-market

Appendix: Time/Cost Benefit Analysis

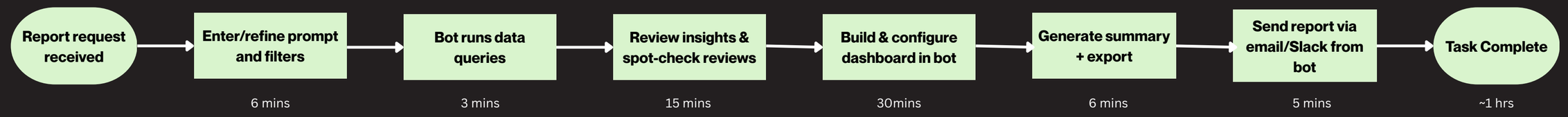
Step	Mins
Clarify scope	20
Pull & join internal data	30
Filter/clean subset (1BR, Brooklyn)	45
Manually read & summarize reviews	80
Manual comp research on Airbnb	120
Build slides/report	100
Email & small follow-ups	20
Total manual time (hours)	**7 hours**

Step	Time (mins)
Enter/refine prompt & filters	6
Bot query runs (8s each × ~5–6 queries, plus overhead)	3
Build & configure dashboard views/filters	15
Review results & spot-check reviews	30
Generate narrative summary + minor edits	6
Export & send via email/Slack	5
Total with bot (hours)	**~1 hour**

"As-Is" Process[4]



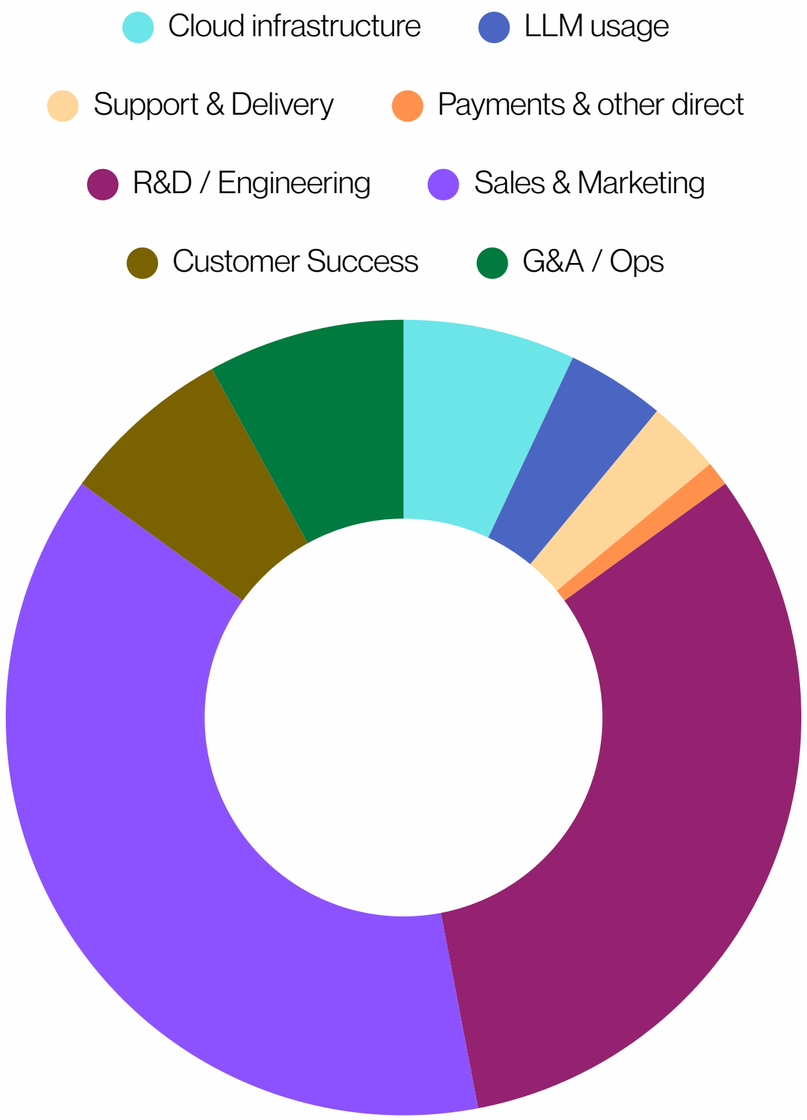
With wtchtwr[4]



Appendix: Business Model

Tier	Listings	Price / month	Price / year
Starter	0–50	$150	$1,800
Growth	51–200	$450	$5,400
Pro	201–500	$900	$10,800
Enterprise	500+	$2,000 (min)	$24,000+

Year	COGS %	OpEx %	Total Cost % of Revenue
Y1	15%	85%	100%
Y2	13%	82%	95%
Y3	12%	80%	92%

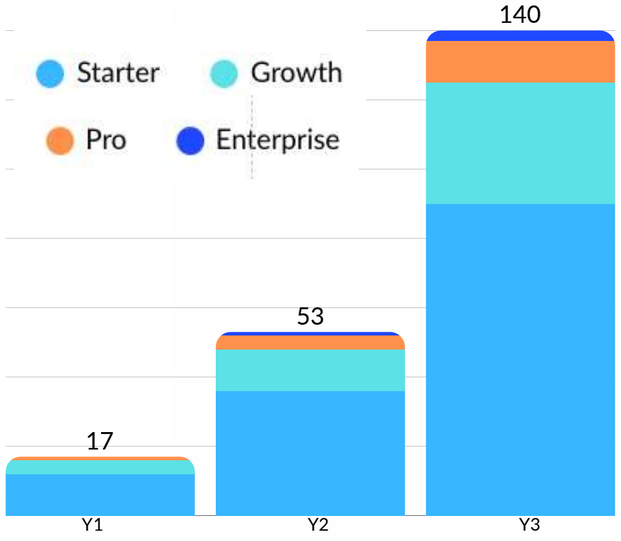


Appendix: Customer Acquisition Strategy

Year	Starter	Growth	Pro	Enterprise	Total Customer
Y1	12	4	1	0	**17**
Y2	36	12	4	1	**53**
Y3	90	35	12	3	**140**

Year	Revenue	Total Costs	Net Profit	Cumulative Cash*
Y0	$0	$0	$0	**$350,000**
Y1	$54,000	$54,000	**$0**	**$350,000**
Y2	$196,800	$186,960	**$9,840**	**$359,840**
Y3	$552,600	$508,392	**$44,208**	**$404,048**

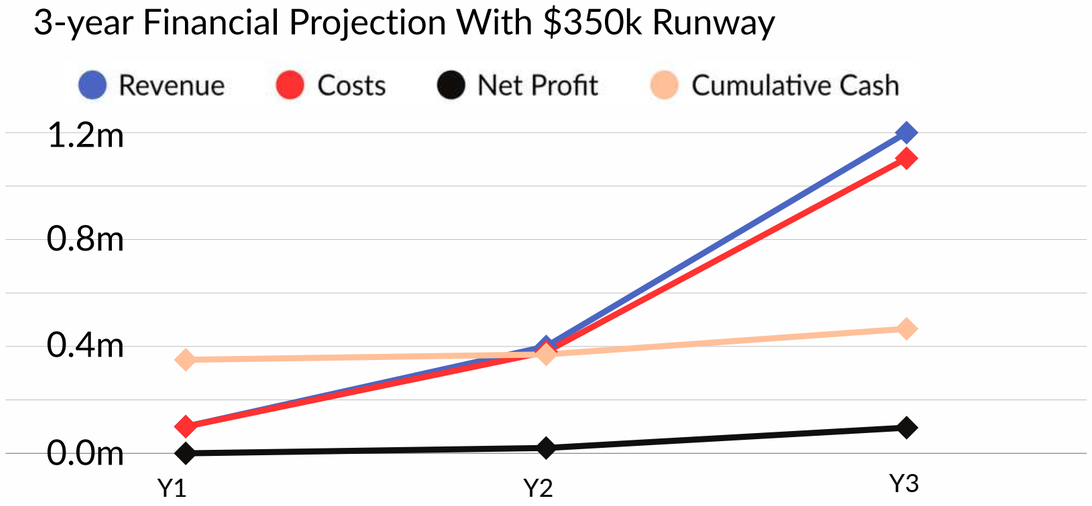
Year	Blended CAC	2-year LTV (gross	LTV/CAC	Payback
Y1	$2,000	$7,500	**3.75x**	~6–7 months
Y2	$2,600	$8,500	**3.27x**	~7–8 months
Y3	$3,200	$9,500	**2.97x**	~8 months



85–88%
Gross Margin

~3-4x
LTV/CAC

<7
months
CAC PAYBACK



Blossom Street Ventures (2024). Percent of SaaS Revenue for R&D, S&M, COGS & G&A https://blossomstreetventures.medium.com/percent-of-saas-revenue-for-r-d-s-m-cogs-and-g-a-8f8cfbe33c2a

Pilehvar, A. (2025). DNSC 6331 – Customer Analytics (course notes)